



AI-driven scientific advertising: ethics, visibility, and research integrity

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ABSTRACT

The integration of AI-driven advertising into scientific discovery platforms like Elicit.com and Consensus.app generates a critical ethical tension: while enhancing visibility and revenue, such advertising risks conflating ad-driven prominence with scholarly merit, threatening research integrity, trust dynamics, and equitable knowledge dissemination. To resolve this conflict, a trust-centric framework grounded in signaling principles and trust-transfer mechanisms is proposed. This framework addresses tensions between paid promotions and traditional credibility signals, such as peer review and citation impact, through integrated strategies: transparency via labeled disclosures and scholarly relevance thresholds to prevent signal conflict; objectivity through algorithmic segregation and third-party audits to sustain trust transfer; and equity employing progressive bidding models and visibility reserves to counteract systemic bias. By redefining credibility assessment in hybrid scientific ecosystems and providing actionable solutions like rigor-weighted ad algorithms, the framework advances marketing scholarship while aligning monetization with academic values. Consequently, marketing emerges as the essential architect of ethical visibility, ensuring advertising amplifies scientific rigor rather than eroding it in algorithmic knowledge economies.

1. Introduction

The pervasive influence of AI-driven advertising has redefined commercial landscapes, with platforms like Google and Meta leveraging algorithmic personalization and behavioral targeting to optimize ad relevance and consumer engagement. Yet, as AI becomes embedded in scientific research tools, such as Elicit.com and Consensus.app, which prioritize citation networks and methodological rigor [1,2], its potential to function as a marketing channel for scholarly knowledge remains a critical blind spot in marketing literature. Critically, as these LLMs move beyond discovery to generate summaries and syntheses of scientific content, the ethical stakes of advertising integration intensify, raising the prospect of promotional influence being baked into the very fabric of AI-generated scholarly narratives. While marketing scholarship rigorously debates ethical dilemmas in consumer-facing AI advertising (e.g., algorithmic bias, ad skepticism), it neglects a pressing question: How do advertising models, proven successful in commercial realms, reshape trust dynamics and credibility assessments in scientific discovery? Unlike traditional digital advertising, where transactional intent is explicit, the integration of paid promotions into academic AI platforms risks conflating ad-driven visibility with scholarly merit, a tension that

threatens the integrity of researcher trust, citation behavior, and knowledge dissemination.

This ethical tension represents a fundamental departure from traditional research dissemination. Historically, scientific visibility has been governed by a slow, reputation-based ecosystem where credibility is signaled through institutional peer review, journal prestige, and citation accrual over time, processes ostensibly insulated from direct payment for placement. The integration of advertising into AI-driven discovery tools constitutes a paradigm shift: it introduces a market-based bidding layer directly into the algorithmic ranking of scientific knowledge. This fusion creates a novel conflict between the slow signals of scholarly merit and the instantaneous signal of financial capability, thereby challenging the foundational norms of scientific credibility in a way traditional publishing does not. The core ethical challenges thus evolve from transparency about author conflicts of interest to transparency about platform governance, algorithmic inputs, and the financial mechanics of visibility itself.

This gap is both glaring and consequential. Foundational marketing theories like signaling theory and trust-transfer theory have long illuminated how users interpret credibility cues in commercial contexts [3, 4]. However, their applicability remains untested in AI-driven scientific

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ecosystems, where credibility is anchored in non-monetary markers like peer review, citation impact, and methodological transparency. For instance, paid promotions in academic platforms may act as conflicting signals: while traditional markers (e.g., high citations) signal rigor, ads risk signaling commercial bias, eroding trust in algorithmic recommendations [5–8]. This oversight leaves platform designers, policy-makers, and researchers ill-equipped to navigate the ethical trade-offs between ad-driven revenue and the sacrosanct norm of scholarly objectivity. As a conceptual theory-building paper, our aim is to construct a detailed, actionable framework that identifies key mechanisms, proposes testable propositions, and establishes a rigorous foundation for future empirical research and platform governance.

Central to this study is the question: How do advertising models in AI-driven scientific platforms enhance or undermine the ethical foundations of scientific discovery, and what marketing frameworks can reconcile ad relevance with research integrity? Grounded in marketing scholarship, this inquiry is refined into three actionable research questions:

- How do paid advertising placements in AI platforms conflict with traditional credibility signals (e.g., peer review, citations), and how does this tension influence researcher trust?
- What role does ad transparency (e.g., labeled disclosures, algorithmic relevance thresholds) play in sustaining trust-transfer dynamics between platforms and users?
- How might inequities in ad allocation, favoring well-funded institutions, distort visibility hierarchies in scientific knowledge markets?

By addressing these questions, the study makes two pivotal contributions to marketing scholarship. Theoretically, it extends signaling and trust-transfer theories to AI-driven scientific ecosystems, revealing how advertising models disrupt credibility assessments in ways distinct from commercial contexts. Practically, it proposes actionable strategies, such as ethical ad-labeling standards and equity-focused bidding models, to align advertising objectives with the ethical imperatives of scientific research. In doing so, this work not only responds to urgent debates about AI's role in knowledge economies but also repositions marketing as a discipline critical to shaping ethical frameworks for emerging hybrid platforms.

2. Literature review

2.1. Advertising ethics in algorithmic systems: lessons from commercial platforms

Marketing scholarship has long grappled with the ethical complexities of algorithmic advertising in commercial ecosystems, where AI-driven systems prioritize engagement and revenue at the expense of equitable visibility. A core tension lies in the interplay between *algorithmic bias* and *ad relevance*: platforms like Google and Meta deploy machine learning to tailor ads, yet these systems often amplify content from dominant market players, sidelining smaller entities and perpetuating inequities [9–11]. For instance, seminal work by Baker & Hawn [12] highlights how algorithmic bias entrenches systemic disparities, even in non-commercial domains like education, by privileging certain voices over others. Similarly, Kordzadeh and Ghasemaghahi [10] synthesize how bias in algorithmic systems arises from skewed training data and opaque decision-making, reinforcing cycles of exclusion. These dynamics mirror challenges in commercial advertising, where well-funded brands dominate visibility regardless of merit, eroding trust in platforms that conflate paid promotions with organic results [11, 13–15].

This erosion of trust is compounded by ad skepticism, a phenomenon where users increasingly question the objectivity of algorithmically curated content. Research indicates that opaque labeling practices, such

as failing to distinguish sponsored posts, undermine credibility, as users struggle to disentangle commercial intent from unbiased recommendations [13,15–18]. Yet, these insights derive from contexts where transactional goals are explicit, and credibility signals (e.g., brand reputation, user reviews) are inherently commercial [19,20]. Scientific discovery platforms, by contrast, operate under norms where visibility hinges on non-monetary markers of rigor [21,22]: peer review, citation impact, and methodological transparency. Here, the integration of advertising models introduces a paradox. While commercial platforms openly prioritize revenue, academic tools risk conflating ad-driven visibility with scholarly merit [23,24], destabilizing the trust researchers place in algorithmic recommendations. For example, undisclosed promotions in a platform like Elicit.com could signal commercial influence, conflicting with traditional credibility cues like citation counts or journal prestige. This dissonance remains unexamined in marketing literature, which has yet to interrogate how ethical frameworks for commercial AI advertising, such as transparency mandates or [25], apply to ecosystems where trust is predicated on perceived objectivity.

The absence of such research represents a critical gap. Foundational theories like signaling theory and trust-transfer theory have illuminated how users interpret credibility cues in transactional environments [3,4], yet their relevance to hybrid platforms, where scholarly rigor intersects with ad-driven revenue, remains untested. This theoretical lacuna is particularly acute regarding how to verify the neutrality of advertising algorithms, a core requirement for trust in scientific contexts. This oversight leaves stakeholders ill-equipped to address ethical dilemmas unique to scientific AI tools, such as whether undisclosed ads erode trust more severely in academic contexts than in commercial ones, or how equity-focused bidding models might mitigate systemic biases in knowledge dissemination. By bridging these gaps, this study advances a marketing-centric lens to evaluate the risks and opportunities of algorithmic advertising in academia, offering theoretical and practical insights absent in current scholarship.

2.2. Algorithmic bias and researcher behavior: trust at the crossroads of visibility and credibility

Algorithmic bias in AI-driven systems is not merely a technical flaw but a profound marketing challenge, reshaping how users perceive, trust, and engage with content [10,23]. In commercial platforms, seminal work by scholars like Sundar [18], Baker and Hawn, [12] has demonstrated that biased algorithms disproportionately amplify dominant narratives, privileging entities with resources while marginalizing others, a dynamic that erodes user trust and perpetuates inequities. These insights, however, derive from ecosystems where transactional goals are explicit, and user behavior is often driven by convenience or brand loyalty [26,27]. Scientific discovery platforms, by contrast, operate within a paradigm where trust hinges on *credibility signals* rooted in scholarly rigor: peer review, citation impact, and methodological transparency [21,22]. Here, algorithmic bias takes on a distinct ethical dimension, as skewed visibility hierarchies risk conflating ad-driven prominence with intellectual merit, destabilizing the very foundations of researcher trust [28,29].

The interplay between algorithmic recommendations and researcher behavior is mediated by two critical forces: *credibility assessment* and *trust heuristics*. Foundational marketing research by [30] elucidates how individuals evaluate information through central or peripheral routes, depending on perceived relevance and credibility. In scientific contexts, researchers rely on central processing, scrutinizing methodological soundness and citation networks, to assess a paper's value [31]. Yet, AI-driven platforms introduce a paradox: algorithmic recommendations, designed to streamline discovery [32], may inadvertently prioritize content optimized for engagement (e.g., keyword density, preprint popularity) over methodological rigor. For instance, a biased algorithm might surface frequently cited papers from well-funded institutions, sidelining novel work from smaller labs, a phenomenon akin to the

"Matthew Effect" in commercial marketing [33,34], where visibility begets further visibility. This creates a self-reinforcing cycle where algorithmic bias distorts researchers' perception of what constitutes credible science, undermining the integrity of citation behavior.

A pivotal, yet unresolved, challenge in this dynamic is the inherent opacity of complex AI systems. The "black box" nature of many algorithms complicates the verification of neutrality, which is paramount for trust in scientific tools [23]. Marketing literature on algorithmic accountability emphasizes that trust is contingent not just on outcomes, but on the ability to scrutinize the process. In scientific ecosystems, where trust is predicated on objectivity, the inability to audit how advertising inputs influence ranking outputs creates a fundamental accountability vacuum. Trust-transfer theory, pioneered by Stewart, further clarifies this dynamic [4]. When users trust a platform, they extend that trust to its recommendations, a principle evident in the widespread adoption of tools like Google Scholar. However, the integration of advertising models into scientific AI systems complicates this transfer. Undisclosed paid promotions, even if algorithmically relevant, introduce noise into credibility assessments. Researchers, conditioned to equate algorithmic outputs with scholarly rigor, may unknowingly conflate *ad-driven visibility* with intellectual authority, eroding trust in both the platform and the scientific process itself [4,17,23,35]. This dissonance mirrors challenges observed in commercial contexts, where [17] work on ad skepticism reveals that users distrust platforms blurring paid and organic content. Yet, in academia, where objectivity is sacrosanct, the stakes are heightened. A single instance of biased or undisclosed advertising could irreparably damage a platform's reputation, as trust in scientific tools is less forgiving than in consumer-facing systems [23].

The ethical implications of this bias extend beyond individual trust to systemic inequities. Well-funded institutions, equipped to bid for ad placements, could dominate algorithmic rankings, replicating the "rich-get-richer" dynamics observed in commercial advertising [26,36]. Early-career researchers or those in underfunded fields, lacking resources to compete, risk invisibility, an outcome antithetical to the egalitarian ideals of scientific progress. This tension underscores a critical gap in marketing scholarship: while algorithmic bias is well-documented in consumer contexts, its manifestation in scientific ecosystems, where credibility signals are non-monetary and trust is non-negotiable, remains unexplored. Bridging this gap requires reimagining marketing frameworks to address the unique ethical contours of academic AI tools, where transparency, equity, and rigor are not optional but imperative.

2.3. Gaps in scientific advertising: underexplored ethical and systemic challenges

While marketing scholarship has meticulously dissected algorithmic advertising in commercial ecosystems, its ethical and operational implications in scientific discovery platforms remain conspicuously underexamined. Three critical gaps persist at the intersection of marketing theory and academic knowledge dissemination:

First, existing research inadequately addresses how paid promotions interact with traditional scholarly credibility signals (e.g., peer review, citation impact). In commercial contexts, ads function as explicit transactional signals, but in scientific platforms, undisclosed promotions risk conflating financial influence with intellectual merit, a conflict untheorized in marketing literature [24,26,36-38]. For instance, while signaling theory explains how brands use endorsements to build trust [3], it does not account for scenarios where paid placements in AI-driven tools (e.g., Elicit.com) compete with peer-reviewed authority, creating credibility dissonance. This gap leaves platforms and policymakers without frameworks to balance ad-driven revenue with the sanctity of scholarly rigor.

Second, marketing scholarship overlooks the systemic inequities inherent to algorithmic visibility hierarchies in academia. Critiques from

digital platform research highlight how digital platforms amplify dominant actors, sidelining marginalized voices, a dynamic mirrored in scientific ecosystems where well-funded institutions could monopolize ad bidding [11,24,39-41]. However, unlike consumer markets, scientific progress relies on egalitarian access to visibility, making inequitable ad allocation ethically untenable. Current literature fails to propose solutions tailored to academic contexts, such as equity-focused pricing models or transparency mandates, which could mitigate bias while preserving revenue streams.

Third, the trust-transfer dynamics unique to scientific AI tools remain unexplored. In commercial platforms, trust hinges on usability and personalization, but in academic settings, trust is predicated on perceived objectivity. Darke and Ritchie's [17] work on ad skepticism and Sundar's [18] MAIN Model reveal that users distrust platforms blurring paid and organic content, yet no studies examine how this skepticism intensifies when objectivity is paramount. For example, researchers may abandon a platform entirely if promoted papers, even algorithmically relevant, are perceived as commercially tainted, eroding trust irreversibly. This gap underscores the need for marketing theories to adapt to contexts where trust is non-negotiable and fragility.

These omissions hinder the development of ethical frameworks for AI-driven scientific advertising. Marketing scholarship must reconcile ad-driven monetization with the non-transactional ethos of academia, addressing how platforms can prioritize relevance without compromising rigor, equity without sacrificing sustainability, and transparency without triggering skepticism. By bridging these gaps, this study advances a novel paradigm for ethical commercialization in knowledge economies, one that aligns marketing's transactional logic with academia's meritocratic ideals.

2.4. Contrasting dissemination paradigms: from traditional gatekeeping to algorithmic visibility

To fully theorize the ethical implications of AI-driven advertising, it is critical to contrast this emergent model with the established norms of traditional scholarly communication. Historically, the dissemination and visibility of research have been governed by a gatekeeping ecosystem centered on institutional reputation and slow, cumulative signals of credibility [42]. This system operates on principles of scholarly meritocracy, where visibility is mediated by journal prestige (impact factors), rigorous peer review, editorial board selection, and the gradual accrual of citations. Ethical transparency in this paradigm has primarily concerned the disclosure of author conflicts of interest, funding sources, and the integrity of the peer-review process itself [43]. The "advertising" within this system is implicit, embedded in the brand value of journals and academic societies, rather than a direct, transactional purchase of prominence.

The rise of digital and social platforms introduced the first major shift, encouraging researchers to engage in "innovative dissemination" through professional networks (e.g., ResearchGate), blogs, and social media to increase reach and impact [42,44]. This practice, while broadening audiences, largely operated in parallel to, and was often informed by, the traditional reputational hierarchy. The integration of paid advertising into scientific discovery platforms (e.g., Elicit.com, Consensus.app) represents a second, more profound paradigm shift. It introduces a real-time, market-based bidding layer *directly into the algorithmic ranking system* that researchers use for discovery. This creates a fundamental tension: the AI's output, which users may perceive as an objective synthesis of scholarly knowledge, can be directly influenced by financial inputs that are invisible by default [45,46].

This shift reconstitutes the core challenges of ethical transparency and integrity. The concern expands from the transparency of the *research origin* (author conflicts) to the transparency of the *discovery pathway* (platform conflicts, algorithmic inputs). As noted in critiques of AI discovery, when an AI agent searches for research, it may treat a rigorously peer-reviewed article identically to a blog post or fabricated paper

unless publishers explicitly make trust signals like peer-review status machine-readable [47,48]. Advertising compounds this by adding a paid signal that the algorithm must weigh against these (often invisible) credibility markers. Consequently, the potential impact on research integrity escalates from individual cases of misconduct to a systemic risk where the very hierarchy of knowledge presented to researchers is shaped by a combination of financial capability and opaque algorithmic optimization, rather than a curated, community-vetted process [49,50]. This contrast underscores the unique stakes of applying advertising models in scientific ecosystems and clarifies why marketing ethics frameworks cannot be transplanted without significant, theory-driven adaptation. It is this very gap, between legacy assumptions of objectivity and the new reality of market-infused algorithms, that the following trust-centric framework is designed to reconcile.

3. Reconciling advertising and research integrity: A trust-centric framework

The ethical integration of paid advertising into AI-driven scientific platforms (e.g., Elicit.com, Consensus.app) demands a marketing-centric resolution to a fundamental tension: how to harness revenue-driven visibility without compromising the scholarly credibility that underpins researcher trust and citation behavior. Unlike commercial ecosystems, where ads openly serve transactional goals [51], scientific platforms risk conflating ad-driven prominence with intellectual merit, threatening the objectivity that defines academic legitimacy [52,53]. To navigate this paradox, a trust-centric framework is proposed, grounded in three interconnected pillars of marketing theory: transparency as credibility signaling [3], objectivity as trust catalyst [4], and equity as systemic imperative [40]. This triad repositions advertising from a disruptor to a compatibility enhancer, ensuring that visibility strategies amplify, rather than undermine, research integrity. The framework's proposed mechanisms are defined in Table 1. By embedding marketing's theoretical rigor into scientific ecosystems, the framework resolves the commercialization-integrity conflict through mechanisms that prioritize trust preservation as the non-negotiable anchor, transforming paid promotions into ethical tools that coexist with academia's meritocratic ethos.

3.1. Pillar 1: transparency as credibility signaling

Transparency functions as the foundational signal that harmonizes

Table 1
Core Mechanisms of the trust-centric framework.

Mechanism	Operational Definition	Theoretical Anchor	Target Metric
Labeled Disclosures	High-contrast visual tags (e.g., "Sponsored", "Promoted") with entity details, distinct from organic results.	Signaling Theory [3]	100 % disclosure rate; user comprehension >90 %.
Dissonance Threshold	Maximum density of paid cues in a result set (e.g., ≤ 20 % of first-page results are ads).	Cognitive Load Theory	Prevents central cue (rigor) overload.
Algorithmic Segregation	Technical isolation of ad-ranking logic from organic relevance algorithms.	Trust-Transfer Theory [4]	Audit-certified data pipeline separation.
Rigor-Weighted Promotion	Ad ranking score = $(\alpha * \text{Rigor_Index}) + (\beta * \text{Log_Scaled_Bid})$, where $\alpha > \beta$ (e.g., 0.7/0.3).	Equity [40]	Quality dominates bid in final ranking.
Visibility Gini Coefficient	Measure of inequality in ad allocation across institutions (0 = perfect equality, 1 = perfect inequality).	Algorithmic Equity	Maintain coefficient below target (e.g., <0.3).

paid promotions with scholarly credibility markers. Drawing on Spence's [3] signaling theory, which explains how observable cues convey unobservable quality, the proposed framework contends that undisclosed ads in scientific platforms create *signal pollution*, eroding trust by blurring commercial intent with algorithmic objectivity. To prevent this, platforms must implement labeled disclosures (e.g., "Sponsored by [Entity]" in high-contrast design) that act as *peripheral cues*, clearly segregating paid placements from organic results [13,15]. Concurrently, relevance thresholds ensure only papers meeting scholarly benchmarks (e.g., peer-reviewed status, minimum citation counts) qualify for promotion, while funding transparency mandates disclosure of sponsor identities (e.g., pharmaceutical affiliations). To prevent signal conflict and user overload, a dissonance threshold must be enforced, limiting the proportion of promoted content in any single search result page (e.g., ≤ 20 %). This threshold is grounded in cognitive load theory, which posits that exceeding a user's capacity to process multiple information channels (central scholarly cues and peripheral ad cues) leads to heuristic processing and decision errors [54]. This approach leverages marketing's understanding of cue hierarchy: by preserving traditional credibility signals (citations, peer review) as *central cues* and ads as *peripheral supplements*, platforms mitigate cognitive dissonance. The outcome is a symbiotic visibility ecosystem where ads enhance reach without compromising trust, a departure from commercial models where signaling often prioritizes profit over clarity.

3.2. Pillar 2: objectivity as trust catalyst

Objectivity serves as the critical mediator that enables trust transfer from platform to promoted content, a dynamic uniquely heightened in scientific contexts. Stewart's [4] trust-transfer theory demonstrates that user trust in a platform extends to its recommendations, but in academia, this transfer hinges on verifiable neutrality, not personalization. To achieve this, platforms must enforce algorithmic segregation, isolating advertising engines from organic ranking systems to prevent bias seepage. Third-party audits by academic consortia (e.g., COPE, FORCE11) must certify algorithmic objectivity. Given the "black box" challenge of complex AI systems, this requires rigorous, independent auditing of both data pipelines and ranking outputs, combined with public disclosure of advertising's impact on visibility diversity, mechanisms elaborated operationally in Section 3.4. These mechanisms align with marketing's emphasis on source credibility: when researchers perceive platforms as impartial arbiters (e.g., through transparency reports), they extend trust to promoted content. Conversely, opaque systems trigger ad skepticism [17], which in scientific ecosystems, where objectivity is sacrosanct, irreparably damages credibility. Unlike commercial platforms (e.g., Amazon), where trust derives from relevance, scientific tools must prove neutrality through structural safeguards, making objectivity the non-negotiable catalyst for ethical ad integration.

3.3. Pillar 3: equity as systemic imperative

Equity counteracts the systemic biases that threaten to replicate commercial "winner-takes-all" dynamics in scientific visibility hierarchies. Noble's [40] seminal work on algorithmic oppression reveals how digital platforms amplify dominant actors, sidelining marginalized voices [55], a risk exacerbated when well-funded institutions monopolize ad bidding. To redress this, we propose progressive bidding models (e.g., 90 % discounts for early-career researchers), visibility reserves (e.g., 30 % ad slots for humanities/underfunded fields), and rigor-weighted promotion that prioritizes methodological quality over bid value. The rigor-weighted algorithm operationalizes this by calculating a composite score for each candidate paper, where the weight for methodological rigor (α) substantially exceeds the weight for the financial bid (β). For instance, a scoring function could be: $\text{Final Score} = (0.7 \times \text{Rigor_Index}) + (0.3 \times \text{Log_Scaled_Bid})$. The *Rigor_Index* itself could be a normalized composite of peer-review status, citation percentile, and scores from

transparency checklists (e.g., TOP Factor, CONSORT). These mechanisms operationalize equity as a measurable marketing construct, using tools like Gini coefficients to audit ad allocation diversity [56,57]. By redistributing visibility, platforms align with academia's egalitarian ideals while addressing Noble's critique that algorithmic systems often entrench inequality. This pillar transcends commercial advertising norms, where equity is often aspirational, by enforcing structural redistribution, ensuring that advertising serves as a tool for inclusive knowledge dissemination rather than exclusionary gatekeeping.

3.4. Operationalizing the framework: governance and ethical safeguards

As established in [Pillar 2](#), independent auditing is essential to verifying algorithmic objectivity. We now specify the operational infrastructure required to implement such audits at scale, together with the accountability and financing mechanisms that render the framework enforceable.

Auditing Infrastructure. Verification of algorithmic neutrality cannot rely on proprietary code inspection. We instead propose behavioral input-output auditing, a method wherein auditors treat the algorithm as a "black box" and assess decisions solely by examining inputs and corresponding outputs [25,58]. This "audit around the computer" approach permits independent verification without source code access [59]. Accreditation should be vested in established scholarly consortia with disciplinary legitimacy and existing infrastructure for ethical oversight: the Center for Open Science (COS), the Committee on Publication Ethics (COPE), and FORCE11. COPE's Emerging AI Dilemmas forum and FORCE11's working group mechanism provide ready platforms for developing discipline-specific audit standards [60,61].

Standardized Audit Protocol. Building on the three audit functions introduced theoretically, we operationalize them as: (1) data pipeline segregation certification, verifying that ad bidding data and organic scholarly metadata reside in technically separated environments; (2) controlled query testing, submitting standardized paper profiles with systematically varied bid values and measuring disparities after controlling for scholarly quality; and (3) differential impact analysis, quantifying visibility inequality introduced by paid promotions, reported as the change in the Visibility Gini Coefficient between organic-only and ad-inclusive conditions [25]. This approach aligns with emerging algorithmic auditing frameworks that prioritize output transparency over model transparency.

Algorithmic Impact Statements. Platforms should publish annual Algorithmic Impact Statements modeled on the TOP Guidelines 2025 framework [62]. Required disclosures shall include: (a) the proportion of result pages exceeding the 20 % dissonance threshold; (b) the Visibility Gini Coefficient for ad impressions, disaggregated by institution type, geographic region, and discipline; and (c) the distribution of the Rigor-Index-to-bid ratio for promoted content. Following TOP's tiered certification model, compliance may progress from disclosure to third-party attestation to full certification of algorithmic weighting integrity.

Redistributive Financing and Pilot Pathway. The equity pillar's progressive bidding discounts (e.g., 90 % fee reductions for early-career researchers and scholars from low-income economies) and visibility reserves require sustainable financing. We propose a *small levy on high-value bid placements*, capped at 5 % of advertising revenue and administered by an independent consortium (e.g., COS or FORCE11), analogous to Crossref's membership fee model [63]. Feasibility should be demonstrated via a controlled disciplinary pilot, for instance, partnering with Consensus.app to deploy rigor-weighted advertising in psychology using the TOP Factor as the initial Rigor_Index, with iterative refinement informed by measured changes in visibility equity and researcher trust.

These governance specifications transform the trust-centric pillars from theoretical constructs into auditable, institutionally anchored standards. They provide platform designers, policymakers, and the research community with a shared vocabulary and replicable blueprint

for reconciling advertising viability with scientific integrity.

3.5. Theoretical synthesis: marketing as architect of ethical visibility

This trust-centric framework advances marketing scholarship by resolving the core tension in AI-driven scientific advertising: the incompatibility of revenue-driven visibility with scholarly integrity. Through its tripartite design, it extends signaling theory to hybrid ecosystems, demonstrating how paid promotions can function as peripheral cues that coexist with, rather than corrupt, non-monetary credibility markers (e.g., citations, peer review) when hierarchical transparency and cognitive load thresholds are enforced [3,54]. It redefines trust-transfer dynamics by establishing provable objectivity, not personalization or engagement metrics, as the essential mediator for ad acceptance in scientific contexts, a paradigm shift from commercial platforms where trust derives from relevance rather than verifiable neutrality [4]. Finally, it operationalizes algorithmic equity from critique to solution, introducing redistributive mechanisms (e.g., rigor-weighted promotion, visibility Gini coefficients) that transform equity from aspirational discourse into measurable practice, grounding the Gini coefficient in established econometric methods for quantifying inequality [56,57] while advancing Noble's [40] critique of algorithmic oppression. Collectively, these pillars reposition marketing from a disruptor to the architect of ethical visibility, a discipline that engineers systems where advertising amplifies rigor rather than revenue, and trust emerges from transparency, neutrality, and fairness. This synthesis not only resolves the commercialization-integrity paradox but also establishes the theoretical foundation for the actionable governance mechanisms specified above and the broader contributions to marketing scholarship elaborated in the next section.

4. Advancing marketing scholarship: ethical visibility in scientific ecosystems

This study fundamentally repositions marketing as the architect of ethical visibility in AI-driven scientific platforms, transforming paid advertising from a potential disruptor of research integrity into a catalyst for equitable knowledge dissemination. Building on the previous theoretical synthesis, the framework advances marketing scholarship by demonstrating how signaling theory, trust-transfer dynamics, and algorithmic equity can be reengineered for objectivity-sensitive scientific contexts, where credibility hinges on peer review and methodological rigor rather than engagement metrics or personalization. These theoretical contributions now translate into actionable pathways for the key stakeholders in scientific knowledge economies.

4.1. From theory to practice: stakeholder-specific pathways

For platform designers (e.g., Elicit.com, Consensus.app), the framework prescribes three concrete interventions. First, platforms must implement tiered disclosure standards, "Sponsored," "Promoted," and "Algorithmic", with distinct visual signatures that render commercial influence immediately discernible to users. Second, they should deploy rigor-weighted algorithms that prioritize methodological quality over bid value, operationalized through composite Rigor_Indices incorporating peer-review status, citation percentiles, and transparency checklist adherence (e.g., CONSORT, PRISMA, TOP Factor), building on established open science frameworks [21,64]. Third, platforms must enforce algorithmic segregation, ensuring that ad-ranking logic operates independently from organic relevance algorithms, with compliance verified through behavioral audits by accredited consortia.

For policymakers and scholarly bodies, the framework provides a governance blueprint. Mandatory transparency reports should document ad allocation diversity using metrics such as the Visibility Gini Coefficient, disaggregated by institution type, geographic region, and discipline. To address systemic inequities, we propose subsidy pools

financed through a small levy on high-value bid placements (capped at 5 % of advertising revenue), administered by independent consortia (e.g., COS, COPE, FORCE11) and used to fund progressive bidding discounts for early-career researchers and scholars from under-resourced institutions. These mechanisms transform critiques of surveillance capitalism into redistributive policy infrastructure [41].

For researchers and academic institutions, the framework necessitates both individual and collective action. At the individual level, trust-driven platform selection, preferring tools with audited algorithmic segregation and published impact statements, creates market incentives for ethical design. At the collective level, institutional libraries and consortia should mandate third-party audit compliance in licensing agreements, transforming diffuse preferences into enforceable procurement requirements. Furthermore, the academic community must develop citation integrity guidelines that encourage verifying a promoted paper's methodological soundness before citation, ensuring that visibility, however gained, does not automatically translate into unearned scholarly credit.

4.2. Implementation challenges and ethical risk mitigation

Translating this framework into practice will encounter significant challenges that require proactive mitigation. The first challenge concerns the auditing of opaque "black box" algorithms. The solution lies in behavioral input-output audits rather than code inspection, but establishing standardized audit protocols and accrediting independent academic consortia to perform them remains an urgent task for the research community. Second, rigor-weighted systems may be susceptible to gaming; institutions could engage in "rigor-washing" by selectively submitting only their highest-quality papers for promotion or by strategically inflating citation metrics. Mitigating this requires dynamic auditing protocols that monitor for anomalous patterns and continuously refine *Rigor_Index* components to incorporate gaming-resistant indicators such as open science badges and replication study outcomes. Third, the economic sustainability of progressive bidding and subsidy pools must be modeled to ensure platforms remain viable while fulfilling equity mandates. Early simulations suggest that a 5 % levy on high-value placements, combined with tiered pricing models, can sustain equity funds without undermining platform revenue, but these projections require real-world validation through controlled pilots.

Addressing these challenges is not peripheral but central to the framework's credibility. It demands cross-disciplinary collaboration between marketing scholars, computer scientists, research policymakers, and platform developers to co-design validation studies, refine governance protocols, and build the institutional infrastructure necessary for ethical visibility at scale.

This work redefines marketing's purpose in knowledge economies: no longer a vector of commercial influence but the steward of scientific trust, engineering visibility systems where rigor is amplified, not auctioned. By enforcing transparency as credibility signaling, objectivity as verifiable infrastructure, and equity as measurable redistribution, marketing emerges as the antidote to "winner-takes-all" visibility hierarchies. Its highest purpose is architectural, to build the governance systems and incentive structures that ensure advertising amplifies underrepresented excellence rather than entrenched power, making genuine scientific progress impossible to ignore.

5. Conclusion

This study has confronted a pivotal tension in the evolution of AI-driven scientific platforms: the integration of paid advertising without compromising the research integrity that underpins scholarly progress. We established that advertising models, while potent for visibility and revenue, risk eroding researcher trust and distorting knowledge dissemination when they conflict with academia's core credibility signals, peer review, citation impact, and methodological rigor. To resolve

this, we proposed a trust-centric framework grounded in marketing theory, transforming advertising from a disruptor into an enabler of ethical visibility. By leveraging Spence's [3] signaling theory, we demonstrated how transparency mechanisms (e.g., labeled disclosures, cognitive load-based dissonance thresholds) position ads as compatible peripheral cues that coexist with scholarly markers. Through Stewart's [4] trust-transfer lens, we redefined objectivity as the non-negotiable anchor for platform credibility, achievable via algorithmic segregation and behavioral third-party audits. Finally, we operationalized Noble's [40] algorithmic equity into actionable mechanisms like rigor-weighted algorithms (e.g., 70/30 quality-to-bid ratios), progressive bidding, and visibility Gini coefficients, countering exclusionary dynamics.

These contributions reposition marketing at the vanguard of scientific integrity. Theoretically, we pioneered three advances: (1) extending signaling theory to hybrid credibility ecosystems with testable propositions, (2) reengineering trust-transfer for objectivity-sensitive contexts around verifiable neutrality, and (3) transforming equity from critique to measurable practice via defined metrics. Practically, we provided detailed implementation pathways: platforms adopt rigor-weighted algorithms and transparency reports; policymakers mandate subsidy pools and audit standards; researchers and institutions enforce trust-driven selection and citation integrity. This framework does not merely reconcile advertising with academia; it reveals marketing as the steward of ethical visibility, architecting systems where rigor is amplified, not auctioned.

5.1. Limitations

This study is conceptual and theory-building; its propositions await empirical validation. First, the proposed dissonance threshold (≤ 20 % ad density) is derived from cognitive load theory applied to peripheral cue processing, and laboratory experiments and field studies are needed to establish the precise boundary across different user populations and disciplinary contexts. Second, the rigor-weighted algorithm assumes the availability of reliable, gaming-resistant *Rigor_Index* metrics. Current indicators (e.g., journal impact factors, citation percentiles) have known limitations and may themselves perpetuate existing biases; future iterations should incorporate more robust open science indicators such as the TOP Factor or transparency checklists (CONSORT, PRISMA). Third, the governance model presumes platforms' willingness to submit to external audits and adopt equity-focused bidding, which may face commercial resistance; incentive structures and regulatory pathways require further investigation. Finally, the framework is tailored to discovery-stage platforms (e.g., Elicit.com, Consensus.app); its transferability to other scientific AI applications (e.g., automated literature review generators, AI-assisted peer review systems) requires contextual adaptation and additional research.

5.2. Future research directions

This conceptual framework opens several avenues for empirical validation and refinement. First, experimental studies should test the proposed dissonance thresholds (e.g., ≤ 20 % ad density) by measuring their impact on cognitive load, trust perceptions, and credibility assessments across different researcher populations and disciplinary contexts. Second, simulation studies can model the ecosystem effects of rigor-weighted algorithms and progressive bidding on visibility equity, exploring how parameter variations (e.g., different α/β weightings) affect outcomes for well-funded versus under-resourced institutions. Third, field experiments partnering with early-adopter platforms (e.g., Consensus.app) should pilot behavioral audit protocols and assess their effect on researcher trust, platform engagement, and revenue stability. Fourth, qualitative research is needed to understand stakeholder resistance and develop effective governance models for audit consortia and subsidy pools. Comparative studies should also investigate how trust erosion and credibility assessments differ between traditional visibility

mechanisms (e.g., citation accrual, journal prestige) and AI-driven paid promotions, examining whether the proposed transparency and equity safeguards can restore trust more effectively in one paradigm than the other. Cross-disciplinary collaborations, between marketing scholars, AI ethicists, computer scientists, and scientific publishers, are essential to co-design these validation studies and the governance blueprints for emerging platforms.

This work challenges marketing to transcend commercialization and embrace a higher purpose: engineering visibility ecosystems where trust is earned through transparency, equity is enforced through design, and excellence becomes impossible to ignore. In an age of algorithmic discovery, marketing's legacy will be defined not by monetization, but by its capacity to safeguard the integrity of knowledge itself.

Declaration of generative AI and AI-Assisted technologies in the writing process

During the preparation of this work, the authors used LLMs tool in order to improve the language and readability of the article. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

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Nasser Bouchareb: Writing – review & editing, Writing – original draft, Formal analysis, Conceptualization. **Ilham Moussaoui:** Writing – review & editing, Formal analysis.

Declaration of competing interest

The author declares no competing interests.

Data availability

No data was used for the research described in the article.

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